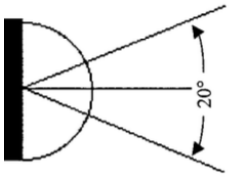
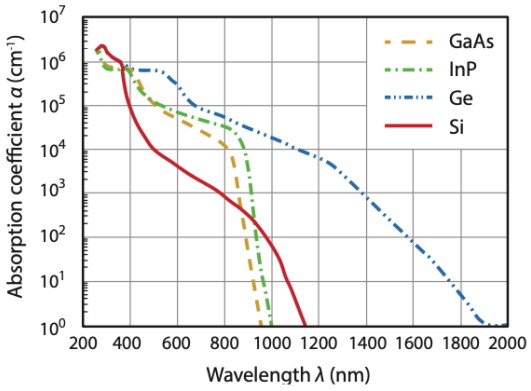
1) Consider a TV remote control design using an infra-red light source with photon energy of 1.5 eV. The light is emitted in a circular cone of 20°, as shown in the figure. The light source power is 0.1 mW. The detector is a silicon photodiode covered with anti-reflection coating. It has absorption of 75% at this wavelength and ξ=1. The detector has dimensions 0.3 cm × 0.3 cm and is placed 1 meter away from the source. Using the plane wave approximation and normal incidence,

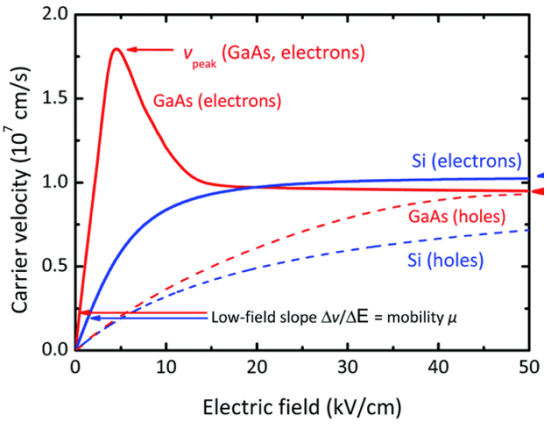


a) Determine the incident electric field amplitude in the light wave.
b) Find the optical power transmitted into the photodiode.
c) What is the responsivity of the photodiode in A/W?
d) What is the photocurrent generated by the photodiode?

2) A GaAs PIN photodiode has an intrinsic region of thickness 0.1 mm. Determine the transit time for the detector and the required reverse-bias voltage. For this photodiodes, determine the efficiency for 800 nm light (take refractive index as 3.6 and ξ=1).

3) Describe a method to determine the optimal load resistance connected across a solar cell for the best fill factor.



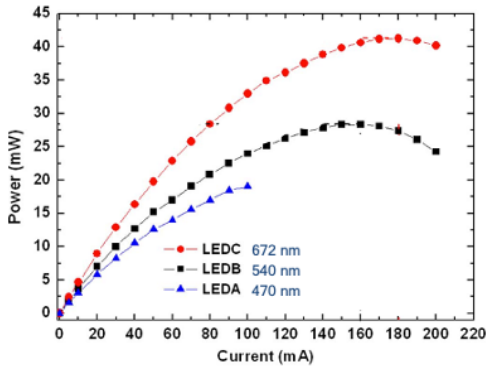


4) Find the peak wavelength, spectral width and escape cone angle of an LED made of $Al_{0.25}Ga_{0.75}As$ at room temperature. Take $\Delta E = 3\ k_B T$. Show all work.

5) What are the two most important reasons why efficient light emission from p-n junctions occurs only in forward bias? Explain.

6) Discuss two methods to solve the problem of absorption of generated photons inside an LED.

7) Find the luminous flux of the LEDA carrying current of 60 mA. Use the given PI curves.



8) Explain the effects of leakage current on the performance of an LED. How can we reduce these effects?

9) Suggest two ways for increasing the modulation bandwidth of an LED. Explain.